



Herbal Therapeutics

Effects of boiled garlic and Shirazi lemon on blood lipids in hyperlipidemic patients: A quasi-experimental study

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ABSTRACT

Aim: As the role of natural compounds in reducing cardiovascular risk factors is of interest to researchers, this study was conducted to determine the effect of boiled garlic and Shirazi lemon on blood lipids.**Methods:** This quasi-experimental study was conducted on patients with hyperlipidemia. Twenty-eight hyperlipidemic patients were randomly selected and evaluated for the effect of boiled garlic and Shirazi lemon on blood lipids. Ten cc of boiled garlic and Shirazi lemon was consumed by the patients in each meal for three weeks. Then, the intervention was repeated after a wash-out period of one week. Blood lipids were measured pre-intervention as well as three and seven weeks later.**Results:** The intervention reduced blood lipid levels significantly ($p < 0.001$). Moreover, mean levels of High Density Lipoprotein decreased three weeks post-intervention and increased significantly post-intervention compared to pre-intervention levels ($p = 0.037$).**Conclusions:** Considering the effects of boiled garlic and Shirazi lemon on the reduction of cholesterol, triglyceride, and Low Density Lipoprotein as well as on the increase of High Density Lipoprotein levels, the use of this decoction is recommended for hyperlipidemia treatment.

1. Introduction

1.1. Background

Cardiovascular diseases (CVDs) are the major causes of mortality and morbidity in the world, unfortunately leading to 17 million deaths every year. It has been estimated that the number will reach 24.8 million worldwide in 2030 (Estruch et al., 2013). Atherosclerotic cardiovascular disease (ASCVD) is defined as a group of clinical diseases mostly caused by endothelial dysfunction based on atherosclerosis, including coronary heart disease, stroke and other various CVDs. These diseases seriously threaten human life and result in high mortality and disability rates all over the world (Xu et al., 2019). A variety of factors influence formation of atherosclerosis, including hypertension, hyperlipidemia, hyperglycemia, obesity and smoking. High levels of blood total Cholesterol

(Chol), Low Density Lipoprotein cholesterol (LDL), Triglycerides (TG), and low level of High Density Lipoprotein cholesterol (HDL) are among the important risk factors. Guidelines to maintain quality of life for people with CVD have focused on decreasing lipids profiles including total and LDL cholesterol level as well as weight control and healthy nutrients use (Deaton et al., 2011; NCEP final report, 2001). These risk factors comprise a majority of key targets for prevention or treatment of ASCVD, among which lipid metabolism disorders, especially abnormal elevation of circulating LDL, are the most important issues. Presently, control of circulating LDL levels has become the most important measure to prevent ASCVD (Ofori-Asenso et al., 2018).

Blood lipid-lowering drugs have numerous side effects that can cause many problems for consumers (Chong et al., 2002). Use of these drugs, especially statins, reduces LDL Cholesterol level, consequently prevents coronary artery disease and reduces mortality. These drugs are mostly

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used in the treatment of hyperlipidemia due to their high serum levels, anti-lipid properties, high tolerance and ease of administration. However, they also have side effects including myopathy, hepatotoxicity, peripheral neuropathy, impaired myocardial contractility and autoimmune diseases (Beltowski et al., 2009). Severe side effects of these drugs have led researchers to seek non-pharmacological treatments of hyperlipidemia.

Dietary factors play a key role in management of hyperlipidemia and CVDs. As healthier foods lemon and garlic are important sources of many different nutrients that are used either raw or cooked. Researchers have reported that citrus fruits intake such as lemon have protective effects against CVD by lowering blood cholesterol level and perhaps by treating hyperlipidemia (Siasos et al., 2013; Benavente Garcia O Castillo et al., 2008). Epidemiologic studies indicate that diets rich in vegetables and fruits are accompanied by a lower risk of mortality (Genkinger et al., 2004). Use of garlic and lemon juice is effective in reducing inflammation status and ultimately cardiovascular parameters (Hussein et al., 2017; Aslani et al., 2016). Garlic is used in various forms such as raw, powdered tablets, or extracted oil. Lipid levels, blood pressure, reactive oxygen species and other cardiovascular risk factors decline under the effect of garlic, as confirmed by several studies (Batsis and Lopez-Jimenez, 2010).

Shabani et al. (2019) have reported that garlic reduces Chol, TG and LDL, and also have recommended to use garlic to reduce blood lipids. In a systematic review, however, Zeng et al. (2013) reported different positive and negative effects of garlic on blood lipids. Results of numerous studies indicated that garlic did not reduce blood lipids (Berthold et al., 1998; Isaacsohn et al., 1998; Simons et al., 1995). In a recent meta-analysis on the effect of garlic on blood lipids, Sun et al. (2018) have concluded that garlic can reduce TC and LDL levels. These different results may be due to the method of use alone or the preparation of a mixture of garlic and lemon. Despite these contradictory results it is recommended that people with high blood lipids should consume more garlic, although more comprehensive research should be carried out in this area.

Lemon is a rich source of vitamin C, citric acid, flavonoids and minerals. It has been used in traditional medicine for many years and has been reported to decrease blood lipid (Naseem et al., 2016).

Fewer side-effects and availability of the natural substances facilitate their use, and considering the studies on the effects of garlic and lemon, and that a high number of studies have been conducted on the relationship between separate use of garlic and lemon and their effects on the reduction of lipid profiles, it is noteworthy to say that very few studies have assessed effects of the boiled garlic and Shirazi lemon in hyperlipidemic patients. When garlic and lemon are boiled, the extract solution is more effective and more active ingredients are obtained compared with other ways of preparation (Zarrabi et al., 2019). In this way, by consuming a small volume of solution the patients receive a large number of useful substances of garlic and lemon. As a result, the present study aimed to investigate effects of boiled garlic and Shirazi lemon in order to manage the increase of lipids in hyperlipidemic patients.

1.2. Objectives

Few studies have been conducted on the effect of a combination of garlic and lemon on blood lipids. For example, Aslani et al. (2016) investigated the combination effect of garlic and lemon on blood LDL levels and reported significant results. However, the formula and preparation method of garlic and lemon in the present study were different from others. Accordingly, the present study aims to investigate the effect of boiled garlic and Shirazi lemon on blood lipids in patients with hyperlipidemia.

2. Materials and methods

2.1. Research design

The present quasi-experimental research involved the manipulation of an independent variable without random assignment of participants to conditions or orders of conditions. This study was a pre- and post-test clinical trial (quasi-experimental) conducted on patients with hyperlipidemia in Gonabad, Iran.

2.2. Subjects and sample size calculation

Based on the study of Mahmoodi et al. (2006) using the mean comparison formula, a sample size of 26 was calculated ($\bar{x}_1 = 126$, $\bar{x}_2 = 122$, $s_1 = 5.5$, $s_2 = 4.8$, $z_{1-\alpha/2} = 1.96$, $z_{1-\beta} = 0.84$) and 36 subjects were considered due to the probability of dropout. From the study population, 123 people met the inclusion criteria. Using a random number table, 36 subjects were selected. Of them, two subjects at the beginning of intervention were dropped out from the study because of their busy schedule and during the study period, two subjects were excluded because of travel and four subjects were removed due to gastric problems. Finally, 28 subjects (14 males, 14 females) remained until the end of the study. The study samples were selected from patients with hyperlipidemia admitted to the Clinic of the 22 Bahman Hospital.

The inclusion criteria were age of over 18 years, subject to hyperlipidemia, resident of Gonabad and willingness to participation. Exclusion criteria were gastrointestinal problems including peptic ulcer, indigestion, gastrointestinal inflammatory diseases, allergy to boiled garlic or/and Shirazi lemon and other chronic diseases.

2.3. Study settings

This study was carried out at the Clinic of the 22 Bahman Hospital affiliated to the Medical University of Gonabad (GMU), Iran, during 2015.

2.4. Intervention

After receiving the laboratory results of HDL, CHOL and TG, 1000 cc of boiled garlic and Shirazi lemon was given to the participants with hyperlipidemia for each stage of the intervention. The solution was obtained by mixing and boiling seven medium-sized Shirazi lemons with peel and 30 garlic glands with four glasses of water (1000 cc) (Saheb-nazar et al., 2016; Abolhassani et al., 2017). Fresh garlic and Shirazi lemon were purchased from a local market in Gonabad. The quality and originality of garlic (No.E1235-FUMH) and Shirazi lemon (No. E1236-FUMH) were confirmed by the Herbarium of Ferdowsi University of Mashhad. Each patient was instructed to consume 10 cc of the solution after each meal for three weeks, then, the participants had a wash-out time for a week, that is, they did not consume boiled garlic and Shirazi lemon during this period. After the fourth week (the wash-out time), they again consumed the garlic and Shirazi lemon solution for another three weeks. In the two stages of the intervention, telephone follow-up was performed on days 6, 12 and 18 for accurate consumption. The patients were visited by a physician each week on a specific day to check for possible complications of boiled garlic and Shirazi lemon. Two patients who reported gastric side effects (heartburn and vomiting) were excluded from the study.

2.5. Measurement

Data were collected by a demographic questionnaire and blood lipid measurements. In order to take blood samples for testing blood lipids in the first stage, the patients were asked to come to the Laboratory of the 22 Bahman Hospital at an appointed time. Blood samples were taken

according to standard methods, then Chol, TG, HDL, and LDL tests were performed for the patients. Blood lipids were measured in a laboratory at the 22 Bahman Hospital using a BIT-3000 analyzer daily monitored for accuracy with Trulab N. To ensure the accuracy of the measurement results, all the tests were performed by one person. Two blood samples from a specific patient were first sent to a laboratory at the 22 Bahman Hospital and then to another laboratory to verify the correlation between the test results.

Three weeks post-intervention, the patients were referred to the Laboratory at the 22 Bahman Hospital at a specified time and blood samples were taken for the second time. Then, the participants had a wash-out time for a week, that is, they did not consume boiled garlic and Shirazi lemon during this period. After this time (the wash-out time), they again consumed the garlic and Shirazi lemon solution for another three weeks. This is the usual method recommended to patients in traditional Iranian medicine (Shiateb, 2014). At the end of the intervention, blood samples were taken for the third time for lipid tests.

2.6. Statistical analysis

Data were analyzed using the SPSS software version 19. First, quantitative variables were examined for normal distribution and then normal quantitative variables were tested using a paired *t*-test and repeated measures ANOVA. The confidence interval in this study was considered to be 95 % and the data were analyzed at the significant level of 0.05.

3. Results

3.1. Characteristics of the study sample

The demographic information of the patients is given in Table 1.

3.2. Overall effect of intervention

The aim of this study was to determine the effect of boiled garlic and Shirazi lemon on blood lipids of hyperlipidemic patients. Overall, all the patients consumed blood lipid lowering tablets. However, laboratory results of pre-intervention blood lipids showed that mean value levels of Chol, TG and LDL were higher than normal, which entirely decreased post-intervention. Further, mean HDL levels increased after the intervention. Altogether, significant reductions of harmful fats were observed after consumption of boiled garlic and Shirazi lemon ($p < 0.001$).

3.3. Effect of intervention at the end of third week

The results of blood lipid measurements at the end of the third week revealed that the mean Chol level decreased from 278.6 mg/dL pre-intervention to 174.28 mg/dL post-intervention. In addition, mean pre-intervention levels of TG (254.35 mg/dL), LDL (205.6 mg/dL) and HDL (40.96 mg/dL) declined to 197.75 mg/dL, 149.39 mg/dL and 39.32

mg/dL, respectively, at the end of the third week. According to the results, all the four types of blood lipids decreased three weeks post-intervention ($p < 0.001$) (Figs. 1 and 2).

3.4. Effect of intervention at the end of intervention

Mean values of Chol (216.78 mg/dL), TG (197.75 mg/dL) and LDL (149.39 mg/dL) at the end of the third week decreased to 174.28, 147.78 and 116.64 mg/dL at the end of the seventh week (the end of the intervention), respectively (Fig. 1). However, the mean post-intervention HDL level (39.32 mg/dL) at the end of the third week increased to 42.67 at the end of the seventh week (Fig. 2). The results showed decrease in Chol, TG and LDL levels compared to the pre-intervention values ($p < 0.001$) (Fig. 1).

Comparison of mean Chol, TG and LDL values pre-intervention, three weeks post-intervention and seven weeks post-intervention revealed that all the three lipids had a decreasing trend from the beginning of the intervention to the end of the intervention, with significant differences ($p < 0.001$). There was a decrease in the HDL level three weeks post-intervention and an increase post-intervention compared to pre-intervention levels. The results of repeated measures ANOVA indicated that these differences were significant ($p = 0.037$) (Table 2).

4. Discussion

One of the most important risk factors for coronary artery disease is hyperlipidemia, which actually initiates the process of atherosclerosis. For this reason, increasing attention has been paid to the control of blood lipids. Nowadays, chemical drugs are used to prevent atherosclerosis, which can cause severe side effects. Therefore, researchers are encouraged to investigate herbal treatment without use of medications.

The overall objective of this study was to determine the effect of boiled garlic and Shirazi lemon on blood lipids of hyperlipidemic patients. The results indicated that the intervention led to a decrease in blood lipids. Blood levels of Chol decreased post-intervention compared to pre-intervention values. Mahmoodi et al. (2006) studied the effects of raw materials on blood biochemical factors in high fat and high blood sugar patients in Rafsanjan and concluded that total Chol levels dropped significantly after 42 days of raw garlic consumption. Abolhassani et al. (2017) also reported a decrease in blood Chol of experimental mice following garlic consumption for three weeks. Rostami et al. (2009) investigated the effect of citrus juice on blood lipids in hypercholesterolemic patients and found significantly reduced levels of serum total Chol after six weeks of continuous consumption of citrus juice along with adherence to the usual regimen for people with high blood lipids. Yaghmae et al. (2009) examined the effect of sour lemon peel essence on blood lipids in adult male Wistar rats and reported that medium and high essence doses significantly reduced Chol concentration. These studies demonstrate that garlic and citrus juice as well as sour lemon peel reduces blood serum lipids. The mixed garlic and sour lemon peel used in the present study confirm the results of previous studies.

Comparison of TG averages at the pre-intervention stage, the end of the third week stage and the post-intervention stage showed that TG decreased significantly from the beginning to the end of the intervention, which is in accord with those of Shabani et al. (2019) who studied the effect of garlic on blood lipids.

Similarly, comparison of LDL averages at the pre-intervention stage, the end of the third week stage and the post-intervention stage pointed out that LDL levels decreased significantly from the beginning to the end of the intervention. In the study of Rostami et al. (2009), citrus juice significantly reduced serum LDL and TC levels. Citrus juice decreased LDL and TC levels while patients were on a normal diet. Daher et al. (2005) examined short- and long-term effects of grapefruit, orange and pineapple consumption on lipids and lipoproteins of mice. They observed that plasma TG decreased with pineapple and grapefruit consumption in the short term, but the amount of total Chol and very

Table 1
Characteristics of the study sample (n = 28).

Variable	Variable levels	n (%)
Gender	Male	14(50)
	Female	14(50)
Age	Mean $\pm$ SD	49.35 $\pm$ 9.32
	Elementary School	15(53.6)
Level of Education	Middle School	5(17.8)
	High school	3(10.8)
	Collegiate	5(17.8)
Earnings	Less than family cost	13(46.43)
	Equal to family cost	8(28.57)
	More than family cost	7(25)

None of the patients was excluded from the study.

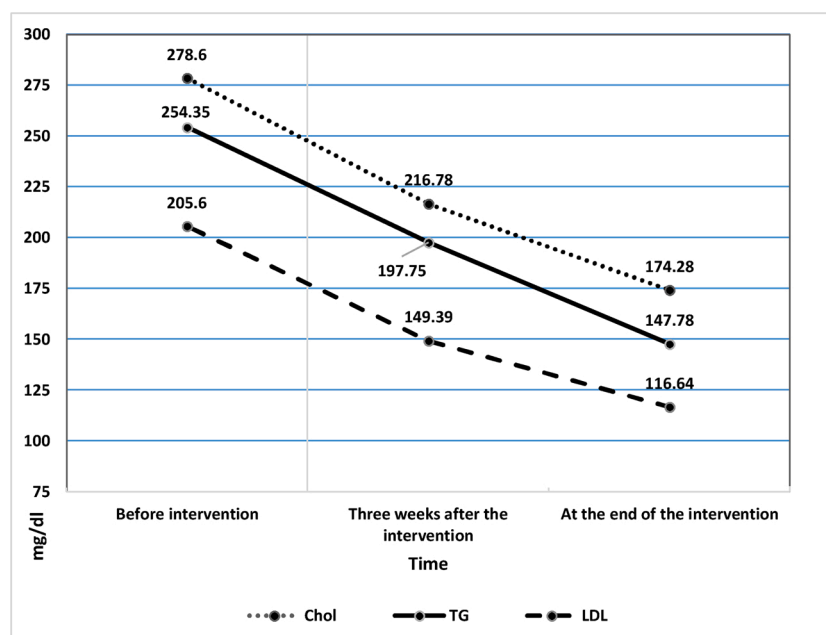


Fig. 1. Comparison of Chol, TG and LDL mean values pre-intervention as well as three and seven weeks later.

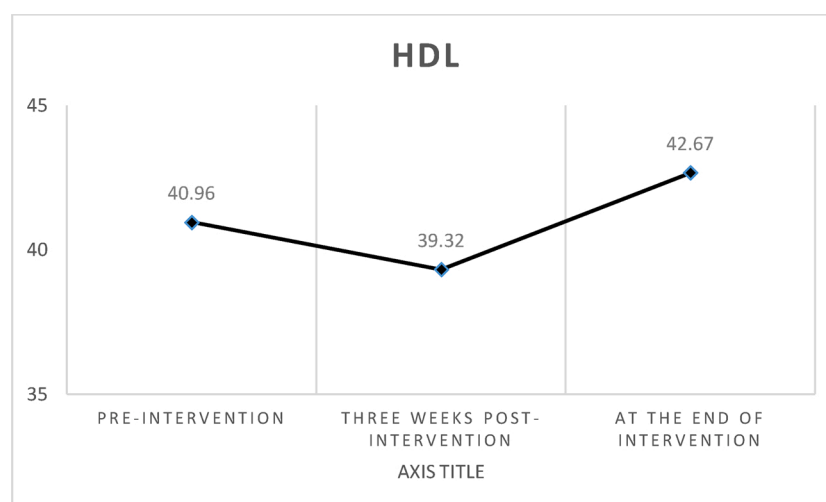


Fig. 2. Mean values of HDL pre-intervention as well as three and seven weeks later.

Table 2

Comparison of mean values of Chol, TG, LDL and HDL.

Time Variable	Time 1 ^a Mean ± SD	Time2 ^b Mean ± SD	Time3 ^c Mean ± SD	RMANOVA
Chol	278.60 ± 14.60	216.78 ± 10.65	174.28 ± 7.93	F = 44.68 P < 0.000
TG	254.35 ± 75.79	197.75 ± 69.59	147.78 ± 49.70	F = 66.80 P < 0.000
LDL	205.6 ± 59.94	149.39 ± 54.85	116.64 ± 45.51	F = 63.81 P < 0.000
HDL	40.96 ± 8.18	39.32 ± 5.51	42.67 ± 4.91	F = 3.50 P = 0.037

^a Pre-intervention.

^b Three weeks post-intervention.

^c At the end of intervention.

low-density lipoprotein cholesterol (VLDL) remained unchanged. Since grapefruit, orange and Shirazi lemon are citrus fruits, triglyceride reduction is one of the results reported by Daher et al. and is comparable with that of the present study.

Comparison of HDL pre-intervention, three weeks post-intervention and four weeks post-intervention indicated significant differences in the averages of the three steps. In a research by Bordia et al. (1998) on patients with coronary heart disease, garlic extract capsules used for three months reduced serum total Chol and TG but increased HDL significantly. Forghani et al. (2001) presented evidence that TG, total Chol, LDL and HDL changed significantly, with an increased HDL as well as a meaningfully decreased LDL, TG and total Chol after vitamin C administration. Since Shirazi lemon, which is rich in vitamin C, was used in the present study, the observed reductions in blood lipids may be linked to vitamin C.

In a study by Ismail et al. (1999), hypolipidemic properties of garlic were compared to those of gemfibrozil in rabbits with hypercholesterolemia and it was detected that garlic could reduce blood Chol levels.

Touhidi and Rahbani (2001) investigated the effect of dried garlic powder tablets on lipids and lipoproteins of 93 hypercholesterolemic patients over a 12-week period. The results showed that total Chol and LDL decreased by 21.2 and 20.4 mg/dl, respectively, in patients fed with garlic powder (2400 mg/day). It was also shown that HDL did not change significantly and TG declined by about 28.1 mg/dl in patients receiving garlic powder (1200 mg/day). In a study by Steiner et al. (1996), administration of garlic extract (7.2 g) for four weeks led to decreased levels of total Chol by 9 %, LDL by 4 % and systolic and diastolic blood pressure by 5.5 %. Mader (1990) reported that 800 mg of garlic for four months reduced Chol by 9 % and TG by 15 %.

A review of studies on effects of garlic, lemon and lemon peel revealed that they could have positive effects on lowering blood lipids. Although negative results have been reported regarding the effect of garlic on blood lipids in mice, studies on humans have shown a positive effect on blood lipids. The use of a combination of garlic, Shirazi lemon and lemon peel in the present study resulted in significant decreases in Chol, TG and LDL concentrations, but a significant increase in HDL levels.

Furthermore, different preparation forms of use for garlic and citrus fruits are suggested. Another difference between this study and other studies is how to prepare the garlic and Shirazi lemon mixture. The formula and preparation method of garlic and Shirazi lemon in the present study were different from those of other studies. In the present study, to prepare boiled garlic and shirazi lemon, initially, 30 garlic cloves were peeled and poured into the mixer with 5 Shirazi lemons with skin that had already been separated from their nuclei, and when all the contents were crushed in the mixer, they were mixed and boiled with 1 L of water (only boiled once). After cooling down, the mixture was filtered, and the contents were kept in a glass and preserved in the refrigerator. But in other studies such as those of Shabani et al. (2019); Aslani et al. (2016); Sun et al. (2018); Touhidi and Rahbani (2001); Yaghmae et al. (2009) and Zeng et al. (2013), the procedure was different. In these studies, a mixture of garlic and lemon without boiling, raw garlic or garlic tablets alone or extract of lemon alone were used. The results of Aliasgharpour (2014) indicated more reduction in triglyceride level than the total and LDL cholesterol levels and no significant reduction was observed in HDL cholesterol level. Also, in Abolhassani's study, the mixture of lemon and garlic only significantly decreased the level of LDL in the rats' blood. Probably, these differences are due to how the garlic and lemon mixture is prepared and method and length of intervention.

5. Limitations and suggestions

One of the limitations of this study was lack of a control group, which decreased the generalizability of the results. Thus, it is suggested to repeat the present study with a control group.

6. Conclusion

Based on the results of this study, it can be concluded that the mixture of boiled garlic with Shirazi lemon is effective on lowering blood lipids of hyperlipidemic individuals. The decoction is highly cost-effective and is favorable to individuals and families. Therefore, it can be concluded that use of this compound in hyperlipidemic individuals can effectively accelerate treatment of this disease and prevent its adverse consequences. Therefore, use of this decoction is strongly recommended in hyperlipidemia treatment.

Ethical considerations

Written informed consent was obtained from all patients participating in the study. Furthermore, this research was approved by Ethics Committee of Gonabad University of Medical Sciences (The Code of Ethics (IR.GMU.REC.1392.81) and the Clinical Trial Registration Code

(IRCTID: IRCT20190317043076N1).

CRediT authorship contribution statement

Hosein Ajamzibad: Conceptualization, Study Design, Data Collection, Statistical Analysis, Data Interpretation, Literature Search, Project administration, Writing - original draft, Manuscript Preparation. **Tahereh Baloochi Beydokhti:** Study Design, Methodology, Supervision, Data Collection, Statistical Analysis, Data Interpretation, Literature Search, Writing - review & editing. **Faeze Mohtasham:** Conceptualization, Study Design, Methodology, Data Collection, Writing - review & editing, Literature Search. **Mahmood Reza Nematollahi:** Study Design, Investigation, Validation, Writing - review & editing.

Declaration of Competing Interest

The authors report no declarations of interest.

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